

Last week
stationarity - Dickey-Fuller test
ACF - PACF in connection with
AR, MA, ARMA, ARIMA processes

Today's lecture

- Holt-Winter, Double Exponential smoothing
- Triple Exponential smoothing
- introduction to Gaussian Mixture Model

Let's say we have a time-series, and we generate a forecast.

$y_1, y_2, \dots, y_t$ } available time-series values.

What is y at time $t+h$?

Let us do exponential smoothing on this data

$$F_{t+1} = \alpha y_t + (1-\alpha)F_t$$

where α = exponential smoothing parameter

leads to exponential moving average

↳ recent values of time more weight than distant values

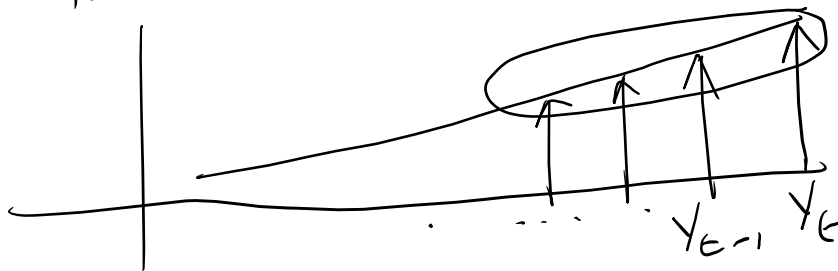
$$F_{t+1} = \alpha y_t + (1-\alpha) \overset{\text{of time}}{\left(\alpha y_{t-1} + (1-\alpha) F_{t-1} \right)}$$

$$\begin{aligned} F_{t+1} &= \alpha y_t + (1-\alpha) \alpha y_{t-1} + (1-\alpha)^2 (F_{t-1}) \\ &= \alpha y_t + (1-\alpha) \alpha y_{t-1} + (1-\alpha)^2 \left(\alpha y_{t-2} + (1-\alpha) \right) \end{aligned}$$

$$\begin{aligned}
 &= \alpha Y_t + (1-\alpha)\alpha Y_{t-1} + (1-\alpha)^2 \alpha Y_{t-2} + (1-\alpha)^3 \alpha Y_{t-3} + \dots \\
 &= \alpha Y_t + (1-\alpha)\alpha Y_{t-1} + (1-\alpha)^2 \alpha Y_{t-2} + (1-\alpha)^3 \alpha Y_{t-3} + \dots \\
 &\quad \text{e.g. (say } \alpha = 0.2 \text{)}
 \end{aligned}$$

Is there any scope for improvement on this?

Suppose there is an increasing trend in the series



if we simply do exponential smoothing
there is a problem

Because of the increasing trend, the forecast at F_{t+1} should be more than the value at Y_t → Simple exponential smoothing will not account for this

Question $\rightarrow$ can we incorporate trend into exponential smoothing?

level variable: $L_t = \alpha Y_t + (1-\alpha) \underbrace{(L_{t-1} + T_{t-1})}_{0 < \alpha < 1}$

trend variable: $T_t = \beta (L_t - L_{t-1}) + (1-\beta) \underline{T_{t-1}}$

forecast: $\hat{Y}_{t+h} = L_t + h T_t$

Holt-Winters Double Exponential Smoothing

$\hat{Y}_{t+1} = L_t + T_t$, $\hat{Y}_{t+2} = L_t + 2T_t$

in regular exponential smoothing, the forecast does not take into account this trend T_t

Holt-Winter, Triple Exponential Smoothing

Level, Trend and Seasonality

- sales have a generally increasing trend, but there is also a seasonal effect
- every December there is a spike in sales over and above what level and trend can account for.

$$\hat{Y}_{t+h} = (L_t + hT_t) S_{t+h} \quad \left. \vphantom{\hat{Y}_{t+h}} \right\} \begin{array}{l} \text{we are trying} \\ \text{to model} \\ \text{the effect of} \\ \text{seasonality} \\ \text{in this fashion} \end{array}$$

In December we might consistently be 20% above the underlying level

$$S_{\text{December}} \approx 1.20$$

Suppose we have M seasons
 $M = 12$ for monthly data

The first year's observations are
 $\underline{Y_1, Y_2, \dots, Y_{12}}$

$$\bar{Y} = \frac{1}{M} \sum_{i=1}^M Y_i$$

Define $S_i = \frac{Y_i}{\bar{Y}}, i = 1, 2, \dots, M$

What is the justification for this?

↳ Why are we dividing by the average.

Assume the model to be this

$$Y_t = L_t S_t$$
 neglect trend for now.

Since we do not know L_t , we use
 the average of the first cycle 12 months to
 estimate $L \approx \bar{Y}$

$$S_t = \frac{Y_t}{\bar{Y}}$$

$$0 < \gamma < 1$$

Recursive updating

$$\underline{S_t} = \gamma \frac{Y_t}{L_t} + (1 - \gamma) \underline{S_{t-M}}$$

↑
current estimate
of seasonality at
time t

previous seasonality value for that same
month $\rightarrow S_{t-M}$

if $t = 13$, we are talking about January
of the second year

$$S_{t-12} = S_{t-M} = S_1$$

it is cycle length dependent $\rightarrow$ cycle length
 $= M = 12$

Initial level:

$$L_{13} = \frac{Y_{(13)}}{S_1}$$

} why is 13 the initial level?

- Data from 1 to 12 is needed to seed the recursion

The first 12 months are used to calculate
 $S_1, S_2, \dots, S_{12}$
" " " " " "

Initial trend:

$$T_{13} = \frac{Y_{13}}{S_1} - \frac{Y_{12}}{S_{12}}$$

Level formula:

$$\underline{L_t} = \alpha \frac{Y_t}{S_{t-M}} + (1-\alpha) \boxed{(L_{t-1} + T_{t-1})}$$

Trend formula

$$T_t = \beta (L_t - L_{t-1}) + \underline{(1-\beta) T_{t-1}}$$

Seasonal factor formula

$$S_t = \gamma \frac{Y_t}{L_t} + (1-\gamma) S_{t-M}$$

The actual forecast:

$$\boxed{\hat{F}_{t+1} = (L_t + T_t) S_{t-M+1}}$$

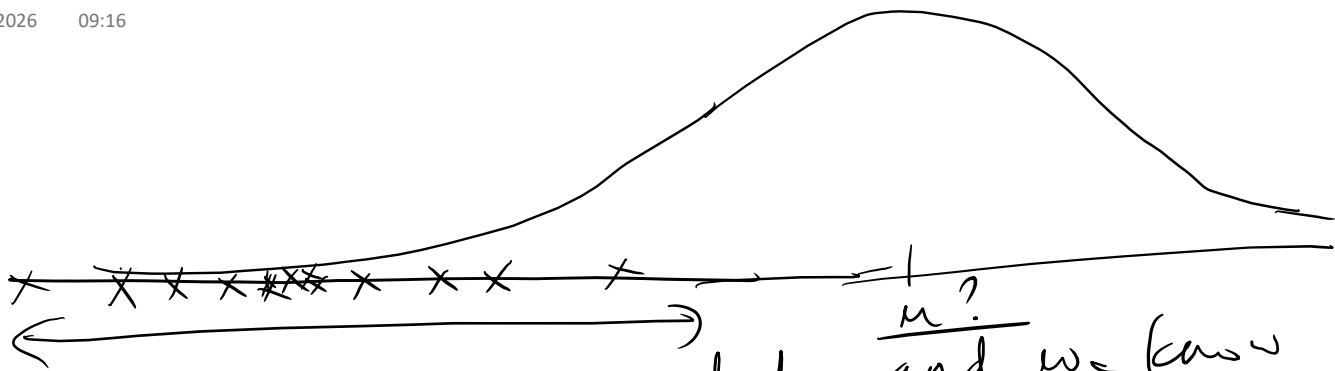
Gaussian mixture model

- how has the given data been
generated?

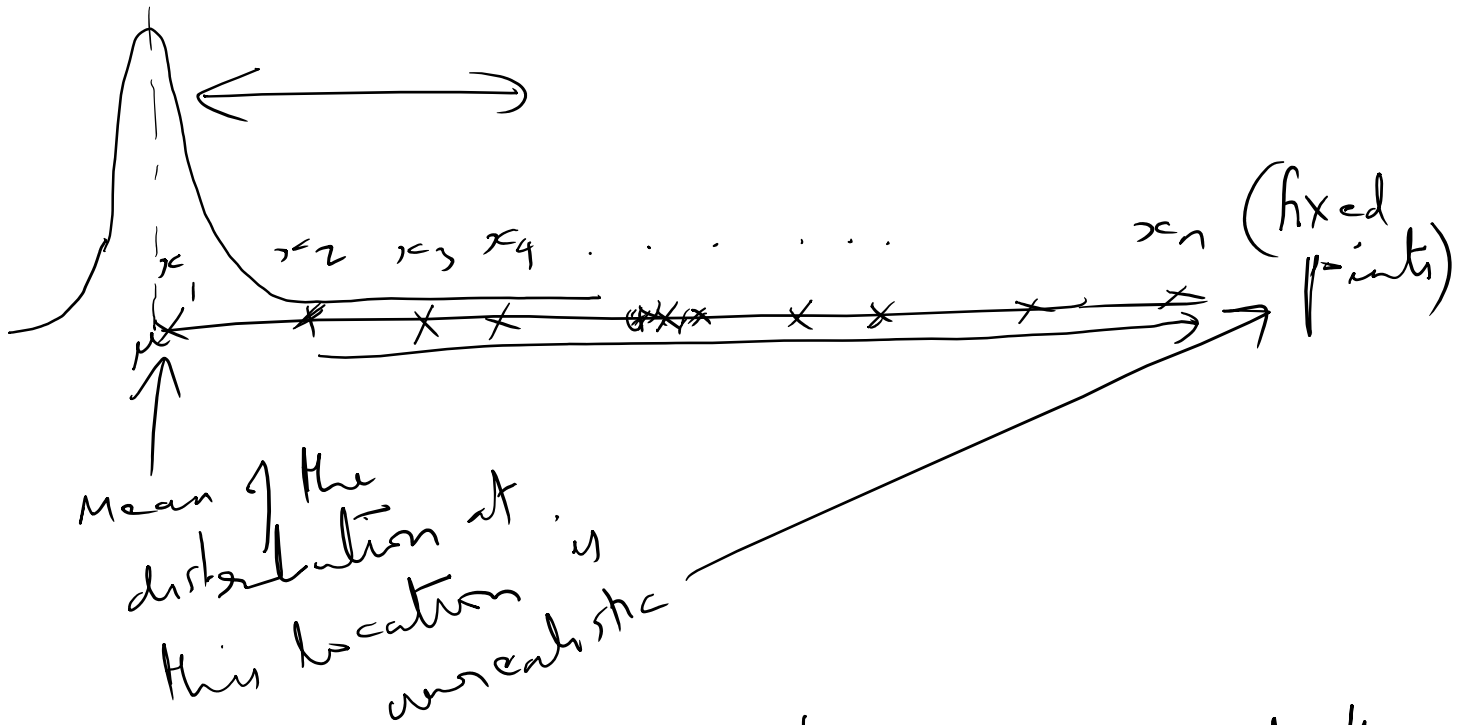
- how do you generate data
that reflects the constraints followed by
the given data?

Can we use the given data to
determine which distribution it
was drawn from?

Once we have the distribution, it is
as good as having the key to generate
new data.



if this is the given data, and we know
that the data is Gaussian, can we say
something about the mean & variance
of the Gaussian?



if the mean is on either extreme end, there is a very small prob for the given data to have been sampled from such a distribution

if our data is between -5 and $+5$ and our mean is at 100 with a sigma of 1

(assume Gaussian dist)

↳ according to the dist, most of

the data will be between 97 and 103

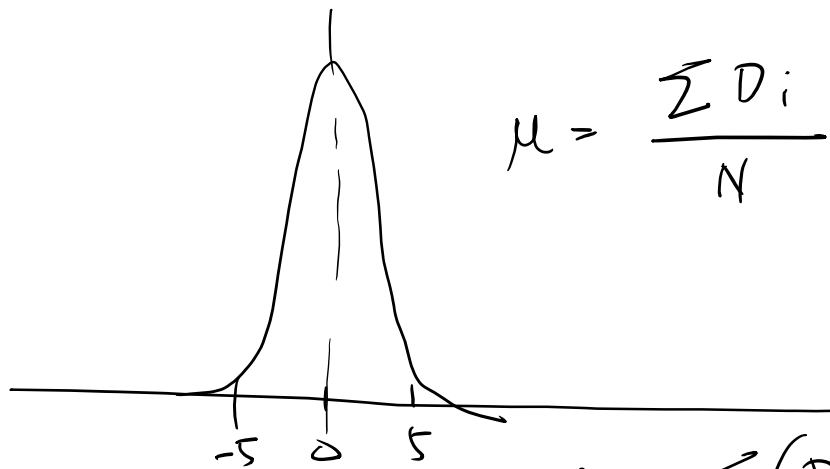
$$\mu - 3\sigma \quad \mu + 3\sigma$$

where $\mu = 100$ and

$$\sigma = 1$$



total prob contained
here is miniscule.



$$\mu = \frac{\sum D_i}{N} \left. \begin{array}{l} \text{estimate of} \\ \text{the mean} \\ \text{of the} \\ \text{data} \end{array} \right\}$$

$$\sigma^2 = \frac{\sum (D_i - \mu)^2}{N}$$

sample mean & sample variance
to get the Gaussian dist
parameters

- set up the likelihood of the data
for a given argument of the
parameters of the Gaussian dist
 $\theta = (\mu, \sigma)$

$$\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_1 - \mu}{\sigma} \right)^2} = P_r(x_1 | \theta)$$

$$\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_2 - \mu}{\sigma} \right)^2} = P_r(x_2 | \theta)$$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_2 - \mu}{\sigma}\right)^2} = Pr(x_2 | \theta)$$

the likelihood of the data

is $Pr(x_1, x_2, \dots, x_n | \theta)$
if all the X_i s are independently generated from the Gaussian dist, we have

$$Pr(x_1, x_2, \dots, x_n | \theta) = Pr(x_1 | \theta) Pr(x_2 | \theta) \dots$$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_1 - \mu}{\sigma}\right)^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_2 - \mu}{\sigma}\right)^2} \dots$$

We have a function in (μ, σ)

$$L(\mu, \sigma) = \left[\frac{1}{\sigma\sqrt{2\pi}} \right]^n e^{-\frac{1}{2}\left(\frac{x_1 - \mu}{\sigma}\right)^2 - \frac{1}{2}\left(\frac{x_2 - \mu}{\sigma}\right)^2 \dots}$$

question is can we maximize this likelihood as a function of μ and σ ?

$$\left. \begin{aligned} \frac{d}{d\mu} [\log L(\mu, \sigma)] &= 0 \\ \frac{d}{d\sigma} [\log L(\mu, \sigma)] &= 0 \end{aligned} \right\} \text{will give formulas for optimal } \mu \text{ and } \sigma$$

[Suppose the given data is generated by multiple Gaussians, and we only know the number of such Gaussians

Finding the best arrangement of such Gaussians is no longer possible with an analytical formula like in the case of a single Gaussian

- EM algorithm (expectation-maximization)

E-step in which we calculate the parameters of the current set of Gaussians

M-step in which we improve the values of the parameters

E-step $\rightarrow$ M-step $\rightarrow$ E-step $\rightarrow$ M-step $\rightarrow$...

how long should we go on?

$$\hat{\mu}_1, \hat{\sigma}_1$$

$$\hat{\mu}_2, \hat{\sigma}_2$$

Suppose this is
the given data
& we are three

$$\hat{\mu}_3, \hat{\sigma}_3$$

Gaussians

E-step $\rightarrow$ M-step $\rightarrow$ E-step $\rightarrow \dots$

Look up the lectures of Andrew Ng for
Gaussian mixture models

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \rightarrow \text{one-dimensional normal dist}$$

$$\frac{1}{|\Sigma|^{D/2} (2\pi)^{D/2}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)} \quad \left[\text{dimension} = D \right]$$

$$\rightarrow \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_x \sigma_y \\ \sigma_x \sigma_y & \sigma_2^2 \end{bmatrix}_{2 \times 2}$$

When $D=2$

Note that the general formula for multiple dimensions becomes what we have for one dimension (set $D=1$)

$$\text{For } D=1 \quad \Sigma = \begin{bmatrix} \sigma^2 \end{bmatrix}_{1 \times 1}$$

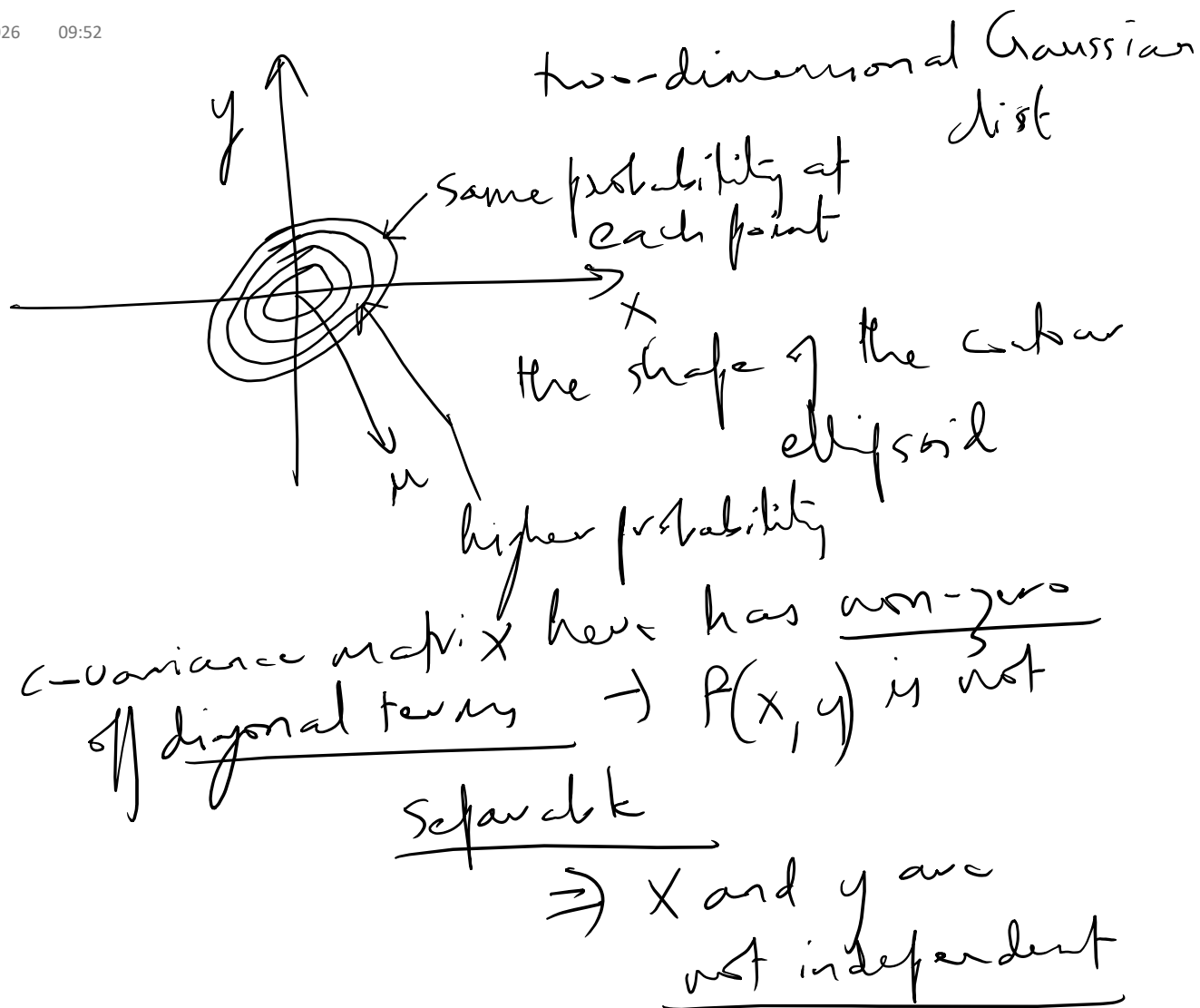
$$\Sigma^{-1} = \frac{1}{\sigma^2} \quad \text{and} \quad \det \Sigma = |\Sigma| = \sigma^2$$

Vector calculus

$$\frac{d}{dx} (x^T \underline{A} x) \rightarrow Ax$$

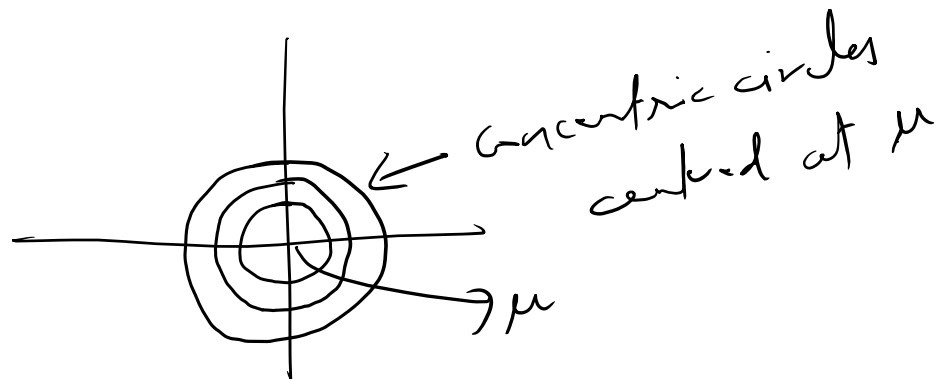
log likelihood will have terms
like $(x - \mu)^T \Sigma^{-1} (x - \mu)$

$$\frac{d}{d\mu} \left[(x - \mu)^T \Sigma^{-1} (x - \mu) \right]$$



When $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$ then the

contour of the joint distribution looks like



Why is $p(x) = \sum_{k=1}^K \pi_k N(x/\mu_k, \Sigma_k)$ a
distribution? $0 < \pi_k < 1$

$0 < p(x) < 1$ for all x

$$\int p(x) dx = 1$$

$$\int \sum_{k=1}^K \pi_k N(x/\mu_k, \Sigma_k) dx$$

$$\underbrace{\pi_1 \int_{-\infty}^{\infty} N(x/\mu_1, \Sigma_1) dx}_{=1} + \pi_2 \int_{-\infty}^{\infty} N(x/\mu_2, \Sigma_2) dx + \dots + \underbrace{\pi_K \int_{-\infty}^{\infty} N(x/\mu_K, \Sigma_K) dx}_{=1}$$

$$\underline{= \pi_1 + \pi_2 + \dots + \pi_K = 1}$$

K-faced die where the k th face
has probability π_k

- ① roll this die and pick the distribution corresponding to the k value on the top
- ② sample the distribution to generate the i th point

The actual parameter vector
 is $\Theta = (\pi_K, \mu_K, \Sigma_K)$

$\uparrow$ $\uparrow$
 mixing K means
 coefficients

K matrices
 each of which
 is $K \times K$

